Supplementary material for the article “Model-comparison effect size indices for logistic and other generalized linear models: Computation and power analysis”

Simulation results tables

The figures in the main text plot the results of Study 1 (recovery of the whole-model R^2 and of a single focal predictor’s eta-squared) and Study 2 (simulation-based required sample size). For Study 1, each Model x N combination in the figures is a curve over 16 population effect size values; the tables below summarize accuracy and power collapsed across those 16 values (mean absolute bias, RMSE, and mean power gap, by model and sample size), rather than reproducing every plotted point – the full per-point results are available in this repository’s `simulations/Data/` folder. For Study 2, which has no sample-size dimension, the tables report the full underlying data.

Study 1a: accuracy of R^2 (collapsed summary of Figure 1)

Table 1: Accuracy of R2 and adjusted R2, collapsed across population R2 (data underlying Figure 1).

Model	N	Mean |Bias| (R2)	RMSE (R2)	Mean |Bias| (Adjusted R2)	RMSE (Adjusted R2)
Gaussian	25	0.088	0.164	0.012	0.156
Gaussian	50	0.042	0.105	0.006	0.101
Gaussian	75	0.028	0.083	0.004	0.080
Gaussian	100	0.021	0.070	0.003	0.069
Logistic	25	0.099	0.177	0.012	0.146
Logistic	50	0.050	0.111	0.006	0.099
Logistic	75	0.032	0.084	0.003	0.078
Logistic	100	0.024	0.070	0.002	0.066
Multinomial	25	0.113	0.156	0.022	0.109
Multinomial	50	0.060	0.095	0.008	0.073
Multinomial	75	0.039	0.070	0.005	0.058
Multinomial	100	0.028	0.057	0.003	0.049
Ordinal	25	0.075	0.139	0.016	0.118
Ordinal	50	0.036	0.086	0.008	0.077
Ordinal	75	0.024	0.066	0.005	0.061
Ordinal	100	0.017	0.055	0.003	0.052

Study 1a: power of the omnibus test (collapsed summary of Figure 3)

Table 2: Mean \|simulated - closed-form\| power for R2 and adjusted R2, collapsed across population R2 (data underlying Figure 3).

Model	N	Mean Power Gap (R2)	Mean Power Gap (Adjusted R2)
Gaussian	25	0.214	0.009
Gaussian	50	0.119	0.007

Table 2: Mean |simulated - closed-form| power for R2 and adjusted R2, collapsed across population R2 (data underlying Figure 3). *(continued)*

Model	N	Mean Power Gap (R2)	Mean Power Gap (Adjusted R2)
Gaussian	75	0.077	0.004
Gaussian	100	0.055	0.002
Logistic	25	0.137	0.027
Logistic	50	0.074	0.010
Logistic	75	0.048	0.006
Logistic	100	0.035	0.003
Multinomial	25	0.221	0.060
Multinomial	50	0.110	0.034
Multinomial	75	0.071	0.023
Multinomial	100	0.051	0.019
Ordinal	25	0.099	0.009
Ordinal	50	0.046	0.003
Ordinal	75	0.030	0.001
Ordinal	100	0.020	0.001

Study 1b: accuracy of eta-squared (collapsed summary of Figure 2)

Table 3: Accuracy of eta-squared and adjusted epsilon-squared, collapsed across population eta-squared (data underlying Figure 2).

Model	N	Mean |Bias| (Eta2)	RMSE (Eta2)	Mean |Bias| (Epsilon2)	RMSE (Epsilon2)
Gaussian	25	0.015	0.129	0.025	0.131
Gaussian	50	0.008	0.091	0.014	0.093
Gaussian	75	0.006	0.076	0.009	0.076
Gaussian	100	0.004	0.065	0.007	0.065
Logistic	25	0.028	0.139	0.012	0.135
Logistic	50	0.018	0.098	0.004	0.096
Logistic	75	0.013	0.078	0.004	0.076
Logistic	100	0.007	0.065	0.002	0.065
Multinomial	25	0.035	0.101	0.028	0.099
Multinomial	50	0.019	0.072	0.009	0.069
Multinomial	75	0.013	0.057	0.006	0.056
Multinomial	100	0.009	0.049	0.005	0.048
Ordinal	25	0.032	0.118	0.013	0.113
Ordinal	50	0.015	0.078	0.005	0.077
Ordinal	75	0.011	0.062	0.004	0.061
Ordinal	100	0.007	0.052	0.003	0.052

Study 1b: power of the single-predictor test (collapsed summary of Figure 4)

Table 4: Mean |simulated - closed-form| power for eta-squared and epsilon-squared, collapsed across population eta-squared (data underlying Figure 4).

Model	N	Mean Power Gap (Eta2)	Mean Power Gap (Epsilon2)
Gaussian	25	0.041	0.038
Gaussian	50	0.030	0.011

Table 4: Mean |simulated - closed-form| power for eta-squared and epsilon-squared, collapsed across population eta-squared (data underlying Figure 4). *(continued)*

Model	N	Mean Power Gap (Eta2)	Mean Power Gap (Epsilon2)
Gaussian	75	0.021	0.005
Gaussian	100	0.014	0.005
Logistic	25	0.056	0.023
Logistic	50	0.026	0.007
Logistic	75	0.016	0.004
Logistic	100	0.010	0.002
Multinomial	25	0.063	0.024
Multinomial	50	0.033	0.007
Multinomial	75	0.019	0.004
Multinomial	100	0.014	0.003
Ordinal	25	0.040	0.012
Ordinal	50	0.017	0.003
Ordinal	75	0.010	0.002
Ordinal	100	0.008	0.002

Study 2: required sample size (data underlying Figure 5)

Table 5: Required sample size for 80% power, omnibus test: simulated vs. closed-form.

Model	Population R2	Required N (Simulated)	Required N (Closed-form)
Logistic	0.025	312	315
Logistic	0.050	156	157
Logistic	0.075	102	105
Logistic	0.100	77	79
Logistic	0.125	60	63
Logistic	0.150	51	52
Logistic	0.175	42	45
Logistic	0.200	38	39
Multinomial	0.025	246	248
Multinomial	0.050	122	124
Multinomial	0.075	80	83
Multinomial	0.100	59	62
Multinomial	0.125	46	50
Multinomial	0.150	38	41
Multinomial	0.175	32	35
Multinomial	0.200	28	31
Ordinal	0.025	198	198
Ordinal	0.050	98	99
Ordinal	0.075	65	66
Ordinal	0.100	49	50
Ordinal	0.125	39	40
Ordinal	0.150	32	33
Ordinal	0.175	27	28
Ordinal	0.200	24	25

Table 6: Required sample size for 80% power, single-predictor test: simulated vs. closed-form.

Model	Population Eta2	Required N (Simulated)	Required N (Closed-form)
Logistic	0.025	226	226
Logistic	0.050	114	113
Logistic	0.075	75	75
Logistic	0.100	56	57
Logistic	0.125	44	45
Logistic	0.150	37	38
Logistic	0.175	32	32
Logistic	0.200	29	28
Multinomial	0.025	176	175
Multinomial	0.050	87	88
Multinomial	0.075	57	58
Multinomial	0.100	43	44
Multinomial	0.125	34	35
Multinomial	0.150	28	29
Multinomial	0.175	25	25
Multinomial	0.200	23	22
Ordinal	0.025	144	143
Ordinal	0.050	73	71
Ordinal	0.075	48	48
Ordinal	0.100	36	36
Ordinal	0.125	29	29
Ordinal	0.150	25	24
Ordinal	0.175	21	20
Ordinal	0.200	19	18

Supplementary code: generating a logistic-regression sample with a known population R-squared

Purpose

This file illustrates, for a single case – a logistic regression whose population R-squared is exactly known in advance – the calibrate-then-sample method used throughout the simulations reported in the paper. The code below is a stripped-down extract of the logistic branch of `sample_by_r2()` from the `Rsimcity` package (https://github.com/mcfanda/RsimCity): input validation, caching, and the branches for the other model families (gaussian, multinomial, ordinal) and for targeting a single predictor's eta-squared rather than the whole model's R-squared have all been removed, so that only the logic essential to this one case remains. All of those variations, their defaults, and their full implementations can be found in the package itself.

The method in three steps

1. Build a numerical-integration grid over the joint distribution of the `k` predictors, using Gauss-Hermite quadrature.
2. Find, by root-finding, the coefficient scale that makes the *population* R-squared – computed exactly on that grid – equal the requested target.
3. Only then draw an actual finite sample of size `n` from the calibrated population.

Code

```
sample_logistic_r2 <- function(n, R2, k = 1, beta = rep(1, k),
                                intercept = 0, x_mean = rep(0, k), x_cov = diag(k),
                                order = 10, seed = NULL) {

  ## Gauss-Hermite quadrature grid over the k predictors
  ## using statmod::gauss.quad.prob(). It returns nodes/weights for a standard normal density,
  ## with weights already summing to 1, so that for any function f, E[f(X)] = sum(w * f(Xgh)).
  ## The grid is then transformed through the Cholesky factor of x_cov and shifted by x_mean, giving qu
  ## for the actual (correlated, possibly non-centered) predictor distribution.
  gh <- statmod::gauss.quad.prob(order, dist = "normal")
  g  <- as.matrix(expand.grid(rep(list(seq_len(order)), k)))
  Z  <- matrix(gh$nodes[g], ncol = k)
  w  <- apply(matrix(gh$weights[g], ncol = k), 1, prod)
  Xgh <- sweep(Z %*% chol(x_cov), 2, x_mean, "+")

  ## Population deviance R-squared, as a function of a coefficient scale `a`
  ## beta = a * beta direction; the model's probability at every grid node gives the
  ## deviance-based R-squared exactly (no simulation is involved in this computation).
  r2_of_a <- function(a) {
    eta  <- intercept + drop(Xgh %*% (a * beta))
    prob <- stats::plogis(eta)

    p    <- cbind(1 - prob, prob)
    p    <- pmax(p, .Machine$double.eps)
    p0   <- colSums(p * w)                      # population marginal probabilities
    D1   <- -2 * sum(w * rowSums(p * log(p)))    # model deviance
    D0   <- -2 * sum(p0 * log(p0))               # null-model deviance
    1 - D1 / D0
  }

  ## ---- 3. Root-find the coefficient scale that produces the target R2 ----------------
  upper <- 1
```

```r
  while (r2_of_a(upper) < R2 && upper < 1e6) upper <- upper * 2
  a <- stats::uniroot(function(a) r2_of_a(a) - R2, c(0, upper))$root
  beta_final <- a * beta

  ## ---- 4. Draw the actual finite sample from the calibrated population ---------------
  if (!is.null(seed)) set.seed(seed)
  X <- sweep(matrix(stats::rnorm(n * k), n, k) %*% chol(x_cov), 2, x_mean, "+")
  colnames(X) <- paste0("x", seq_len(k))
  eta <- intercept + drop(X %*% beta_final)
  y   <- stats::rbinom(n, 1, stats::plogis(eta))

  data.frame(y = y, X)
}
```

Example: a single predictor, R-squared = .10

```r
dat <- sample_logistic_r2(n = 2000, R2 = .10, seed = 1)
mod <- glm(y ~ x1, data = dat, family = binomial())
1 - mod$deviance / mod$null.deviance   # sample deviance R2, close to the .10 target
```

```
## [1] 0.09601619
```

Example: three correlated predictors, R-squared = .20

```r
Sigma <- matrix(c(1, .3, .2,
                  .3, 1, .4,
                  .2, .4, 1), nrow = 3)

dat3 <- sample_logistic_r2(n = 5000, R2 = .20, k = 3, x_cov = Sigma, seed = 1)
mod3 <- glm(y ~ x1 + x2 + x3, data = dat3, family = binomial())
1 - mod3$deviance / mod3$null.deviance   # close to the .20 target
```

```
## [1] 0.1916439
```